

The risks that the Mercury astronauts faced were soon clear enough. The program was being rushed through – the original goal was for a first manned flight in mid-1960 – but rocket technology remained dangerously unreliable. The plan was for a suborbital flight using a Redstone booster to be followed by an orbital flight using a more powerful Atlas launcher. But in July 1960 an Atlas test-fired with a Mercury capsule on top blew up spectacularly just after launch, and the following November, the Redstone-Mercury combination proved a dud, lifting barely 6in (15cm) off its launch pad before settling back to rest.

### Pipped at the post

There was thus no end of relief at NASA when on January 31, 1961, a chimpanzee called Ham was successfully sent into space strapped into a Mercury capsule. The event went far from smoothly – Ham eventually splashed down 130 miles (210km) off target and nearly drowned before being picked up. Nor were the proud and skillful Mercury pilot-astronauts pleased to be reminded that the flight they were training for had been successfully carried out by a monkey. But NASA was ready for a human launch in the spring, only pushing it back from March to May in order to iron out the glitches revealed by Ham's space adventure. It was a small delay, but it let the Soviets win.

In complete contrast to the United States, where the glare of publicity surrounding the space program was dazzling, the Soviet Union proceeded in secrecy. The appearance was of an unbroken series of triumphs, each emerging without prior announcement – with a suddenness that in practice enhanced their impact. And the triumphs were real. In 1959 they won the first Moon race with their Luna spacecraft, crashing Luna 2 on to the Moon's surface and sending Luna 3 around the Moon to take the first photos of its invisible dark side. But there were also

setbacks that were hidden from the world. In October 1960 an attempt to send an unmanned spacecraft to Mars failed, and this was followed by the worst single disaster of the space age, when a new rocket booster on its launch pad at Baikonur unexpectedly ignited while being examined by senior officers, engineers, and technicians. Over 100 people died, engulfed in flames, yet the event went totally unreported.

Like the Americans, the Soviets were hell-bent on being first to send a man into space, and selected a team of trainee cosmonauts from their pool of air force fighter pilots. The 20 Soviet cosmonauts were significantly younger than their American counterparts, ranging in age from 24 to 34. In contrast to the instant fame of the Mercury 7, the cosmonauts were unknown to the public, their very existence a secret. As a result, they were dispensable. In March 1961, cosmonaut Valentin Bondarenko, aged 24, carried out a series of tests that involved living in a pressurized oxygen chamber. He had an electric hot plate in the chamber for cooking. At the end of the tests, he carelessly tossed a piece of alcohol-soaked cotton onto the hot plate, causing the oxygen-soaked atmosphere to burst into flames. Badly burned, Bondarenko died shortly afterward.

Because of its secrecy, such an event was no hitch in the progress of the Soviet manned space program. From May 1960, the Soviets had begun test launches of their Vostok space capsule, sending a veritable menagerie of animals into orbit, including dogs, mice, rats, frogs, and literal, rather than figurative, guinea pigs. By April 1961 they were ready for the real thing. Only three weeks after the death of Bondarenko, Gagarin went into orbit and became, for a time, the most famous man in the world.

There was no doubt of the political motive behind the Soviet space effort, or of its effectiveness. Across the developing world, where new nations were becoming independent of colonial rule every year, leaders and their peoples saw the Soviet Union as the world's leading

### YURI GAGARIN

**YURI ALEXEYEVICH GAGARIN** (1934–68) was born on a collective farm near Smolensk. As a child he survived the Nazi occupation during World War II, as well as the harshness of life under Stalin's regime. In the 1950s he went to technical college, and was expected to become a factory supervisor. But instead he found a place on an air force training course and in 1957 graduated as a pilot officer. In March 1960, at the age of 26, he was selected to train as a cosmonaut. This was no passport to glamour. Recently married and starting a family – his second child was born a month before the first space flight – Gagarin lived in a modest Moscow apartment and commuted to cosmonaut training in the city suburbs.

Becoming a Soviet hero turned his life into a round of public ceremonies and official foreign visits. Gagarin always longed to return to flying and to space. He had his wish in 1968, resuming cosmonaut and jet fighter training. But in March of that year he was killed when his MiG15UTI trainer crashed.



#### CHARMING HERO

*Raised to the rank of colonel in the Soviet Air Force, Yuri Gagarin proved an excellent ambassador for his country, displaying confident charm in his public role as a Soviet hero.*



#### TARGETING THE MOON

*On May 25, 1961, President Kennedy told Congress that the USA should commit itself to "landing a man on the Moon and returning him safely to Earth" before the end of the decade.*

"This is the new age of exploration; space is our great New Frontier."

**JOHN F. KENNEDY**

WHILE US PRESIDENTIAL CANDIDATE IN 1960

technological power and Soviet-style communism as the way of the future. Losing the space race was causing real damage to the global interests of the United States at a time of intense Cold War confrontation – this was the year of the failed Bay of Pigs invasion of Cuba and the building of the Berlin Wall.

Once Gagarin had won the race to be first in space, the only way the Americans could hope for victory was by raising the stakes – by setting a more ambitious target, distant enough to give the Americans time to catch up with and overtake the Soviets. The new US president, John F. Kennedy, decided to challenge the Soviet Union to a race to put a man on the Moon. His